Sahaj Bhadja

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Summary

Graduate software engineer with 2+ years building **Python/Java/C++ applications, data platforms, and cloud infrastructure**; experienced in distributed systems, ETL, performance optimization, observability, CI/CD, financial workflows, and collaborative delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **distributed Python services and data infrastructure** across FastAPI, PostgreSQL, SQL Server, and cloud storage, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Engineered resilient asynchronous processing for 1,000+ concurrent jobs with structured states, validation, and failure isolation, increasing throughput 3x while sustaining 99%+ reliability across 5,000+ monthly workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **full-stack React and Python applications** for uploads, processing, history, and analytics, serving 150+ users and 10,000+ daily requests while maintaining stable performance during peak workloads.
- Integrated multiple AI services behind adaptable APIs and owned features through design, implementation, testing, code review, deployment, and production monitoring, reducing feedback turnaround 60% through pragmatic automation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed **cross-platform Java and Python microservices** using OOP, SQL, caching, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, and data consistency across operating environments.
- Collaborated with Microsoft and GitHub engineers across global Agile teams while maintaining 20+ unit and integration tests, Jenkins pipelines, deployment workflows, technical documentation, and production failure reports.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **scalable React and Node.js production platform services** across eight microservices, securely synchronizing 2M+ records through REST APIs, WebSockets, Azure SQL, authentication, permissions, analytics, and operational reporting.
- Automated Docker and Jenkins delivery workflows while monitoring services through Sentry, Azure Monitor, and Kibana, achieving 95%+ build validation and reducing deployment time 50% across production releases.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Developed **large-scale financial applications and data pipelines** for pricing, payments, refunds, and reporting, integrating Flask, Firebase, Databricks, and Spark across 50,000+ records with sub-200-millisecond API responses.
- Mapped and documented 12+ system integrations, implementing secure webhooks, real-time synchronization, and cross-platform workflows while reducing manual refund handling 70% and supporting reliable customer-facing financial transaction operations.

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, Bash, Shell
Software: Data structures, algorithms, OOP, concurrency, distributed systems, REST APIs, microservices, performance optimization
Data: Data engineering, PostgreSQL, SQL Server, MySQL, MongoDB, Firebase, ETL pipelines, Databricks, Spark, data validation
Infrastructure: Linux, AWS, Azure, Docker, Kubernetes, cloud storage, Jenkins, GitHub Actions, CI/CD
Reliability: Sentry, Azure Monitor, Kibana, logging, monitoring, automated testing, debugging, Git, code reviews, Agile

Projects

CPU Scheduling Simulator Live | *C++, React.js, Operating Systems*

- Implemented and benchmarked **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case testing, and dynamic performance metrics across 1,000+ process combinations, reducing average turnaround time 25%.

FinNexus Live | *Kubernetes, MongoDB, WebSockets*

- Built a **distributed financial platform** supporting 50,000+ users and 1M+ transactions with secure bank integrations, real-time updates, MongoDB persistence, advanced filtering, and Kubernetes-based scalable deployment automation.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Enhanced a **cross-platform data application** across Java and TypeScript modules, supporting 10+ databases with schema exploration, query generation, validation, Docker, Maven, Node.js, and reusable Linux-compatible development workflows.